

# REINFORCEMENT LEARNING TASK 1

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### PART-1

1. Financial Trading: Reinforcement Learning is used in algorithmic trading to optimise buy, sell, and hold decisions by learning from market conditions and maximising long-term investment returns.
2. Traffic Signal Control: RL helps optimise traffic light timings by adapting to real-time traffic flow, reducing congestion, minimising waiting times, and improving road efficiency.
3. Supply Chain and Inventory Management: RL is applied to optimise inventory levels, warehouse operations, demand forecasting, and logistics, helping businesses reduce costs while maintaining product availability.
4. Game Playing: RL enables AI agents to learn optimal strategies in games such as Chess, Go, Data 2, and StarCraft II through continuous interaction with the game environment.
5. Cloud Resource Allocation: RL is used in cloud computing to dynamically allocate computing resources, schedule workloads, balance server loads, and reduce energy consumption while maintaining performance.

### PART-2

| Item                 | Paper - 1                                                                                          | Paper - 2                                                                                                               |
|----------------------|----------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------|
| Application          | Traffic Signal Control                                                                             | Cloud Resource Allocation                                                                                               |
| Paper Title          | <b>Traffic-R1: Reinforced LLMs Bring Human-Like Reasoning to Traffic Signal Control Systems</b>    | <b>A Hybrid RL–GA–LSTM–AE Framework for Energy-Aware and SLA-Driven Task Scheduling in Cloud Computing Environments</b> |
| Authors              | Xingchen Zou, Yuhao Yang, Zheng Chen, Xixuan Hao, Yiqi Chen, Chao Huang, Yuxuan Liang              | Bommineni Narsimhulu, T. Sampath Kumar                                                                                  |
| Year                 | 2026                                                                                               | 2026                                                                                                                    |
| Journal / Conference | Proceedings of the 64th Annual Meeting of the Association for Computational Linguistics (ACL 2026) | <i>Scientific Reports</i> (Nature Portfolio)                                                                            |

| Item              | Paper - 1                                                                                                                                                                                                                                                                                                                                                                                                                 | Paper - 2                                                                                                                                                                                                                                                                                                                 |
|-------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Main Contribution | Introduces <b>Traffic-R1</b> , a 3B-parameter reinforced large language model that combines reinforcement learning with human-like reasoning for intelligent traffic signal control. The model generalizes to new road networks, handles emergency incidents, supports multi-intersection coordination, and reduces average traffic queue lengths by over <b>5%</b> while serving more than <b>55,000 drivers daily</b> . | Proposes a hybrid <b>RL-GA-LSTM-AE</b> framework for intelligent cloud task scheduling. The model combines Reinforcement Learning, Genetic Algorithms, and an LSTM Autoencoder to optimize task allocation, reduce energy consumption, improve SLA compliance, and enhance CPU utilization in dynamic cloud environments. |
| RL Algorithm      | <b>Group Relative Policy Optimization (GRPO)</b> with a two-stage Reinforcement Learning framework (offline human-informed RL and online RL).                                                                                                                                                                                                                                                                             | <b>Reinforcement Learning (RL)</b> integrated with a <b>Genetic Algorithm (GA)</b> and <b>LSTM Autoencoder (LSTM-AE)</b> for adaptive scheduling and resource allocation.                                                                                                                                                 |